

Word problems on motion with two speeds

A journey in two parts, and the current or wind that changes the speed.

LESSON 123–125

3 × 40 MIN

MATEMATIKA 6, §26

S7 12 RATES

LESSON CLOCK

25'

30'

35'

25'

5

Two stages	25 min
Setting out a table	30 min
Downstream and upstream	35 min
Harder problems	25 min
Homework	5 min

LEARNING OBJECTIVES

- Handle a journey made in two stages at different speeds.
- Use downstream and upstream speeds.
- Set out the working in a table.
- Check each stage separately as well as the whole.

TEXTBOOK

National	pp. 359–368
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.7

01

Explanation

Two stages

Work each stage separately, then add the distances and add the times. Never add the speeds.

STAGE	SPEED	TIME	DISTANCE
first	60 km/h	2 h	120 km
second	80 km/h	3 h	240 km
total	—	5 h	360 km

SPEEDS ARE NEVER ADDED

The two stages above do not make a journey at 140 km/h. Distances add, times add; a speed is a ratio between them and has to be worked out at the end.

Setting out a table

PROBLEM	TABLE	ANSWER
2 h at 60 then 3 h at 80	$120 + 240$	360 km
90 km at 45 then 60 km at 60	$2 + 1$	3 h
1.5 h at 50 then 30 min at 70	$75 + 35$	110 km

THREE COLUMNS, ONE ROW PER STAGE

Speed, time, distance — with a totals row underneath. The table does the organising, and the arithmetic becomes obvious.

Downstream and upstream

downstream = $v + u$ upstream = $v - u$

with v the speed of the boat in still water and u the speed of the current.

BOAT	CURRENT	DOWNSTREAM	UPSTREAM
16	4	20	12
18	3	21	15
10	2	12	8

THE SAME IDEA WORKS FOR WIND

An aeroplane with a tailwind flies at $v + u$ and against it at $v - u$. The two speeds add and subtract exactly as with a river.

Harder problems

PROBLEM	WORKING	ANSWER
a boat at 16 in still water, current 4: 60 km downstream	$60 \div 20$	3 h
the same boat coming back	$60 \div 12$	5 h
the whole trip	$3 + 5$	8 h
a boat covering 20 km down in 1 h and 12 km up in 1 h	$(20 + 12) \div 2$	boat 16, current 4

HALF THE SUM AND HALF THE DIFFERENCE

Given the two speeds, the boat's own speed is their average and the current is half their difference.
Grade 7 will write that as a system of two equations.

02 Worked examples

EXAMPLE 1 A car drives 2 hours at 60 km/h and then 3 hours at 80 km/h. Find the total distance.

- 1 First stage: 120 km.
- 2 Second stage: 240 km.
- 3 $120 + 240 = 360$ km.
The speeds are not added.

ANSWER

360 km

EXAMPLE 2 A boat travels at 16 km/h in still water. The current is 4 km/h. How long to go 60 km downstream and back?

① Downstream: 20 km/h, so 3 h.

② Upstream: 12 km/h, so 5 h.

③ $3 + 5 = 8$ h.
Not $60 \div 16$ doubled.

ANSWER

8 hours

EXAMPLE 3 A boat covers 20 km downstream and 12 km upstream, each in one hour. Find the speed of the boat and of the current.

① Downstream speed 20, upstream 12.

② Boat: $(20 + 12) \div 2 = 16$ km/h.

③ Current: $(20 - 12) \div 2 = 4$ km/h.
Check: $16 + 4 = 20$ ✓

ANSWER

Boat 16, current 4

03 Interactive model

Draw the table on the board and fill it in with the class before any arithmetic; the organisation is what makes these problems easy.

Two stages, two speeds

2 h at 60 then 3 h at 80: the distance is:

140

360

280

700

The total time is:

5 h

6 h

2.5 h

140 h

Adding the two speeds gives:

the average speed

nothing useful

the total

the distance

A boat at 16 with a 4 current goes downstream at:

And upstream at:

60 km downstream at 20 takes:

And back upstream:

Down 20, up 12: the current is:

4

8

16

32

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Stage of a journey	Yo'l bosqichi	Участок пути
Downstream	Oqim bo'ylab	По течению
Upstream	Oqimga qarshi	Против течения
Current	Oqim	Течение
Still water	Turg'un suv	Стоячая вода
Total distance	Umumiy masofa	Общее расстояние
Total time	Umumiy vaqt	Общее время
Table of working	Jadval	Таблица

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In a two-stage journey you add:

the speeds

the distances and the times

the speeds and times

nothing

Downstream speed is:

$v - u$

$v + u$

vu

v

Upstream speed is:

$v - u$

$v + u$

$\frac{v}{u}$

u

A boat at 18 with a 3 current goes upstream at:

15

21

18

6

Given both speeds, the boat's own speed is:

their sum

their average

their difference

the larger

And the current is:

their average

half their difference

their difference

the smaller

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 2 h at 60 then 3 h at 80: the distance

ANSWER 360 km

2. And the total time

ANSWER 5 h

3. A boat at 16, current 4: downstream

ANSWER 20 km/h

4. And upstream

ANSWER 12 km/h

5. A boat at 18, current 3: downstream

ANSWER 21 km/h

6. And upstream

ANSWER 15 km/h

7. 60 km at 20 km/h

ANSWER 3 h

Medium · the standard question

7 PROBLEMS

1. 90 km at 45 then 60 km at 60: the time

ANSWER 3 h

2. 1.5 h at 50 then 30 min at 70

ANSWER 110 km

3. A boat, 60 km down at 20 and back at 12

ANSWER 8 h

4. Down 20, up 12: the boat

ANSWER 16 km/h

5. And the current

ANSWER 4 km/h

6. Down 24, up 16: the boat and current

ANSWER 20 and 4

7. A plane at 500 with a 50 tailwind

ANSWER 550 km/h

Hard · stretch and reason

7 PROBLEMS

1. A boat at 12 in still water, current 3: 45 km each way

ANSWER 8 h

2. A car does 120 km at 60 and 120 km at 40: the total time

ANSWER 5 h

3. A plane flies 1 800 km out at 600 and back at 450

ANSWER 7 h

4. A boat takes 2 h down and 3 h up for the same 36 km

ANSWER Boat 15, current 3

5. A journey of 300 km, half at 50 and half at 75

ANSWER 5 h

6. Why can the two speeds not be added?

ANSWER A speed is a ratio, not an amount

7. A boat's downstream speed is twice its upstream speed and the current is 3: the boat

ANSWER 9 km/h

07 Homework

Homework — 5 tasks

Draw the three-column table before writing any arithmetic.

1. A car drives 3 h at 70 km/h then 2 h at 90. Find the distance and the total time.
2. A boat travels at 14 km/h in still water with a current of 2. Find its speed each way.
3. How long does that boat take to go 48 km downstream and back?
4. A boat covers 30 km downstream in 1.5 h and 30 km upstream in 3 h. Find the boat and the current.

5. A journey of 180 km is half at 45 km/h and half at 90 . Find the total time.